

AI-Powered Mobility Assistant for Visually Impaired

hardware/hardware_main.py

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# =====
# AI-Powered Mobility Assistant
# Hardware Main Configuration
# Raspberry Pi Hardware Layer
# =====
# -----
# IMPORTS
# -----
import serial
import cv2
import pyttsx3
from picamera2 import Picamera2
import RPi.GPIO as GPIO
import time
import os
# -----
# GPIO CONFIGURATION
# -----
GPIO.setmode(GPIO.BCM)
# Example GPIO Pins
GPS_TX = 14
GPS_RX = 15
GPIO.setup(GPS_TX, GPIO.OUT)
GPIO.setup(GPS_RX, GPIO.IN)
print("GPIO Initialized Successfully")
# -----
# CAMERA INITIALIZATION
# -----
picam2 = Picamera2()
picam2.configure(
    picam2.create_preview_configuration(
        main={"format": "RGB888", "size": (640, 480)}
    )
)
picam2.start()
print("Camera Started Successfully")
# -----
# GPS INITIALIZATION
# -----
try:
    gps = serial.Serial(
        '/dev/ttyS0',
        baudrate=9600,
        timeout=1
    )
    print("GPS Module Connected")
except:
    print("GPS Module Not Detected")
# -----
# AUDIO INITIALIZATION
# -----
engine = pyttsx3.init()
engine.setProperty('rate', 150)
engine.setProperty('volume', 1.0)
print("Audio System Ready")
```

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# -----
# SPEECH FUNCTION
# -----
def speak(message):
    print("Speaking:", message)
    engine.say(message)
    engine.runAndWait()
# -----
# CAMERA TEST FUNCTION
# -----
def test_camera():
    while True:
        frame = picam2.capture_array()
        cv2.imshow("Pi Camera Feed", frame)
        key = cv2.waitKey(1)
        if key == 27:
            break
    cv2.destroyAllWindows()
# -----
# GPS READER FUNCTION
# -----
def read_gps():
    try:
        data = gps.readline().decode(
            'utf-8',
            errors='ignore'
        )
        return data
    except:
        return "GPS Data Not Available"
# -----
# HARDWARE STATUS FUNCTION
# -----
def hardware_status():
    print("\n==== HARDWARE STATUS =====")
    print("Camera   : ACTIVE")
    print("GPS       : ACTIVE")
    print("Speaker   : ACTIVE")
    print("GPIO      : ACTIVE")
# -----
# AUTO START FUNCTION
# -----
def auto_start():
    print("Starting AI Mobility Assistant...")
    os.system(
        "python3 /home/pi/AI-Mobility-Assistant/software/main.py"
    )
# -----
# MAIN HARDWARE LOOP
# -----
if __name__ == "__main__":
    hardware_status()
    speak("Hardware system initialized successfully")
    while True:
        gps_data = read_gps()
        print("GPS:", gps_data)
        frame = picam2.capture_array()
        cv2.imshow("Hardware Camera Feed", frame)
        key = cv2.waitKey(1)
        # ESC key
        if key == 27:

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        break
    # S key -> Start AI software
    if key == ord('s'):
        auto_start()
cv2.destroyAllWindows()
GPIO.cleanup()
print("Hardware System Shutdown")
```